Health and Disease

Week 8

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Today

Between 1850 and 1950 American life expectancy at birth rose by about 30 years. Two of this week's readings ask what produced the gain and the third asks what happens when one input to it is destroyed.

1. **Two theoretical frames.** Disease control as a public good the state must supply and trust as a belief that one piece of news can move.

2. **Institutions.** Troesken on why the richest and freest country in the world had a higher smallpox rate than Ceylon, a poor British colony.

3. **Infrastructure.** Ferrie and Troesken on what moving Chicago's water intake offshore did to deaths from diseases that water does not carry.

4. **Trust.** Alsan and Wanamaker on what the Public Health Service's 40-year deception of 600 men in Alabama did to the health of older Black men, most of all near Macon County.

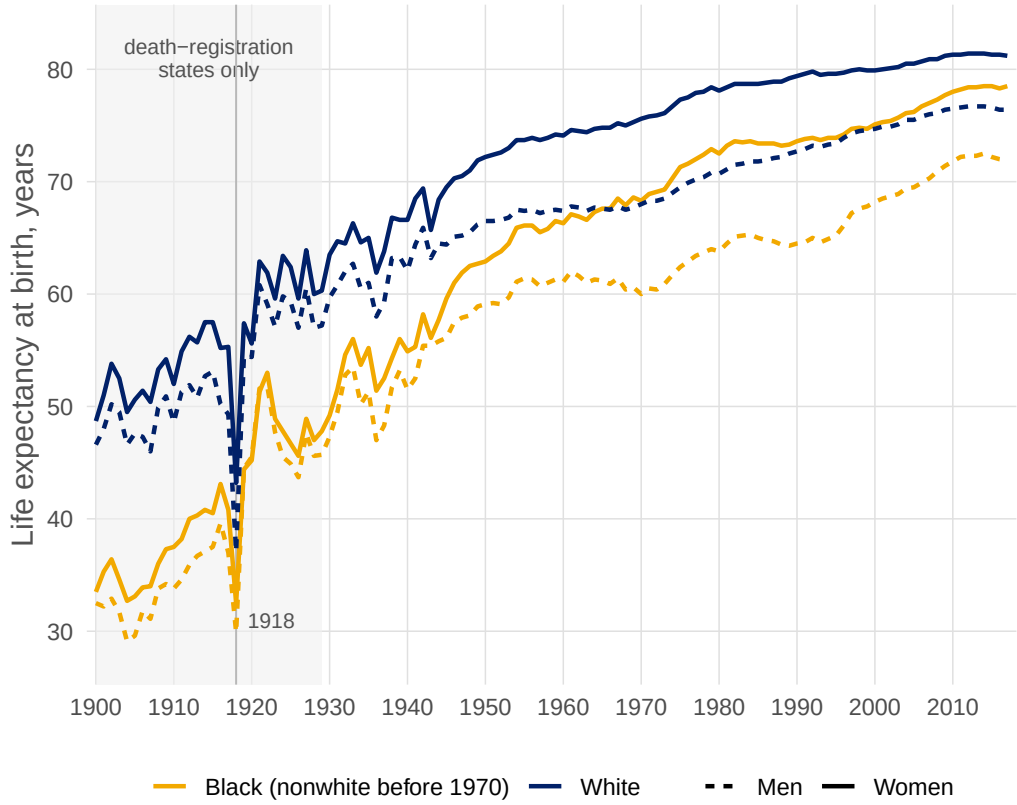
Are democracies better at controlling disease? And how much does trust in the doctor matter for whether medicine gets used?

The mortality transition

Life expectancy at birth, 1850–2017

	Whites	Nonwhites
1850	39.5	23.0
1900	49.6	33.0
1950	69.0	60.8

- Measured nonwhite life expectancy nearly doubled between 1900 and 1950, under Jim Crow. White voters backed water mains into Black neighborhoods for fear typhoid would spread (p. 11).
- After 1950 the rate of improvement fell to between a third and a half of what it had been.



Table, Troesken (2015), p. 10. Chart, NCHS United States Life Tables 2017, Table 19, via `scripts/R/wk08_life_expectancy.R`. Before 1970 the Black series is the nonwhite population. The 1918 dip is the influenza pandemic. The 1850 values are estimates and the 1900 values cover only the death-registration states, which held few Black Americans. NCHS gives 47.6 for whites in 1900 where Troesken prints 49.6.

Diarrheal and respiratory diseases

Disease	Cause	Spread and victims
Diarrheal, spread through sewage in water and food		
Typhoid fever	the bacterium *Salmonella* Typhi	Weeks of fever. Killed 5–10 percent of those infected, many of them young adults.
Cholera infantum	no single germ, a period name for summer diarrhea in infants	Spoiled milk and dirty water in hot months. Infants died of dehydration. It is distinct from epidemic cholera.
Dysentery	*Shigella* bacteria or amoebae	Bloody diarrhea. Deadliest for young children and the old.
Respiratory, spread through the air		
Tuberculosis	the bacterium *Mycobacterium tuberculosis*	Coughing in crowded housing. A slow lung disease, called consumption, that killed young adults.
Influenza	a virus	Seasonal outbreaks. Deadliest for infants and the old, often through the pneumonia that followed.
Bronchitis	inflammation of the airways after infection or from smoke	A broad label for respiratory deaths of infants and the old.
Pneumonia	bacteria, chiefly pneumococcus, or viruses	Infection of the lungs. Often the final cause in people weakened by another disease.

Doctors recorded causes of death from symptoms, before laboratory tests were routine, so the categories overlapped. Ferrie and Troesken (2008), p. 1, list the causes. The descriptions are standard medical summaries.

What people died of

Chicago before 1880

- More than half of all deaths were of children under five.
- The leading causes were diarrheal diseases, among them typhoid fever, cholera infantum, and dysentery.
- Respiratory diseases followed, among them tuberculosis, influenza, bronchitis, and pneumonia.

Chicago in 1925

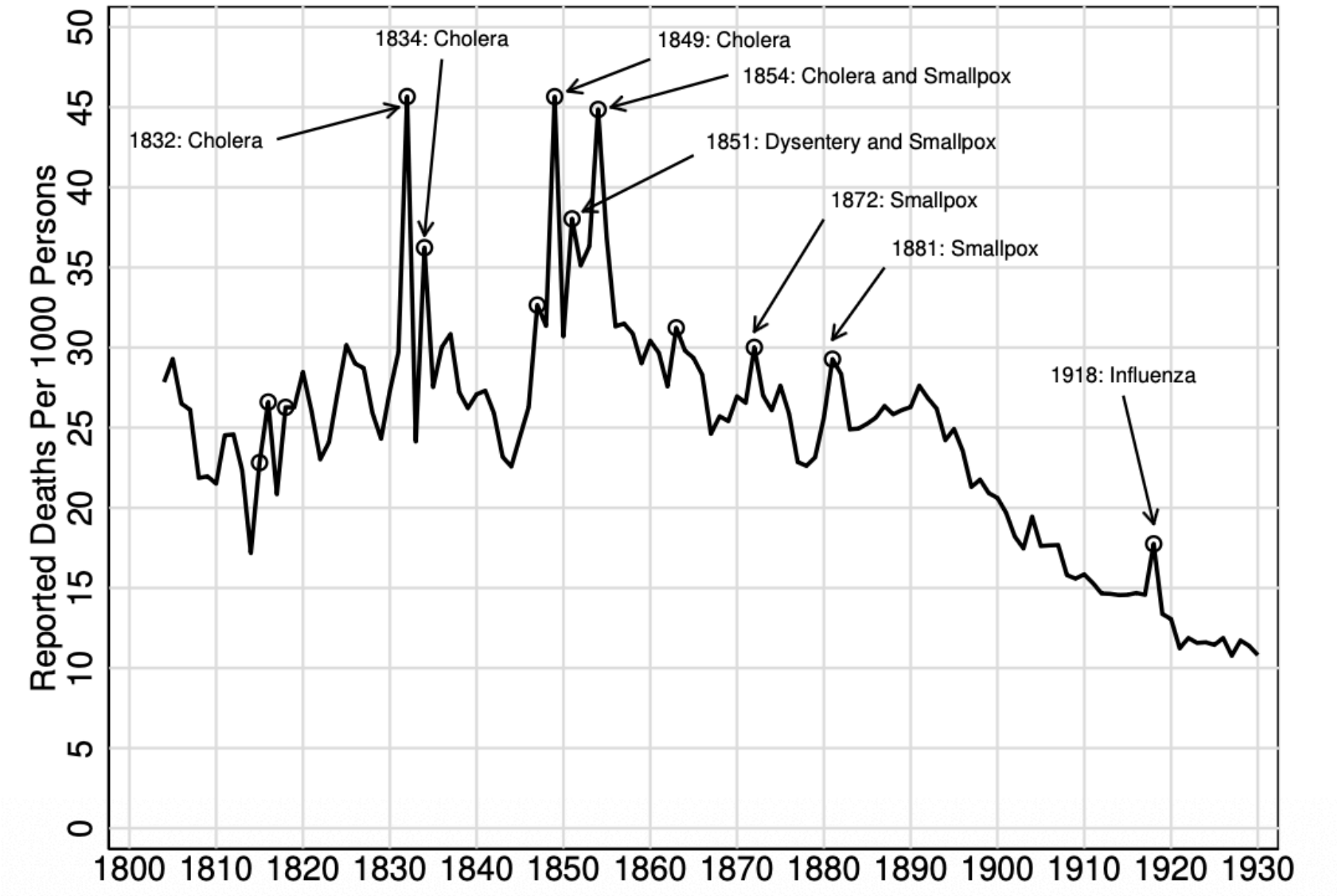
- Under a quarter of deaths were of children under five.
- Typhoid was largely gone.
- Heart disease and cancer were overtaking respiratory disease as the leading causes.

- Ferrie and Troesken report that Boston, Philadelphia, New York, and other large cities made the same transition over the same years and that smaller cities followed a decade or two later.

- They also argue that Chicago's disease environment in 1880, AIDS and industrial pollution aside, matched the disease environment of the developing world in the 2000s, when they wrote.

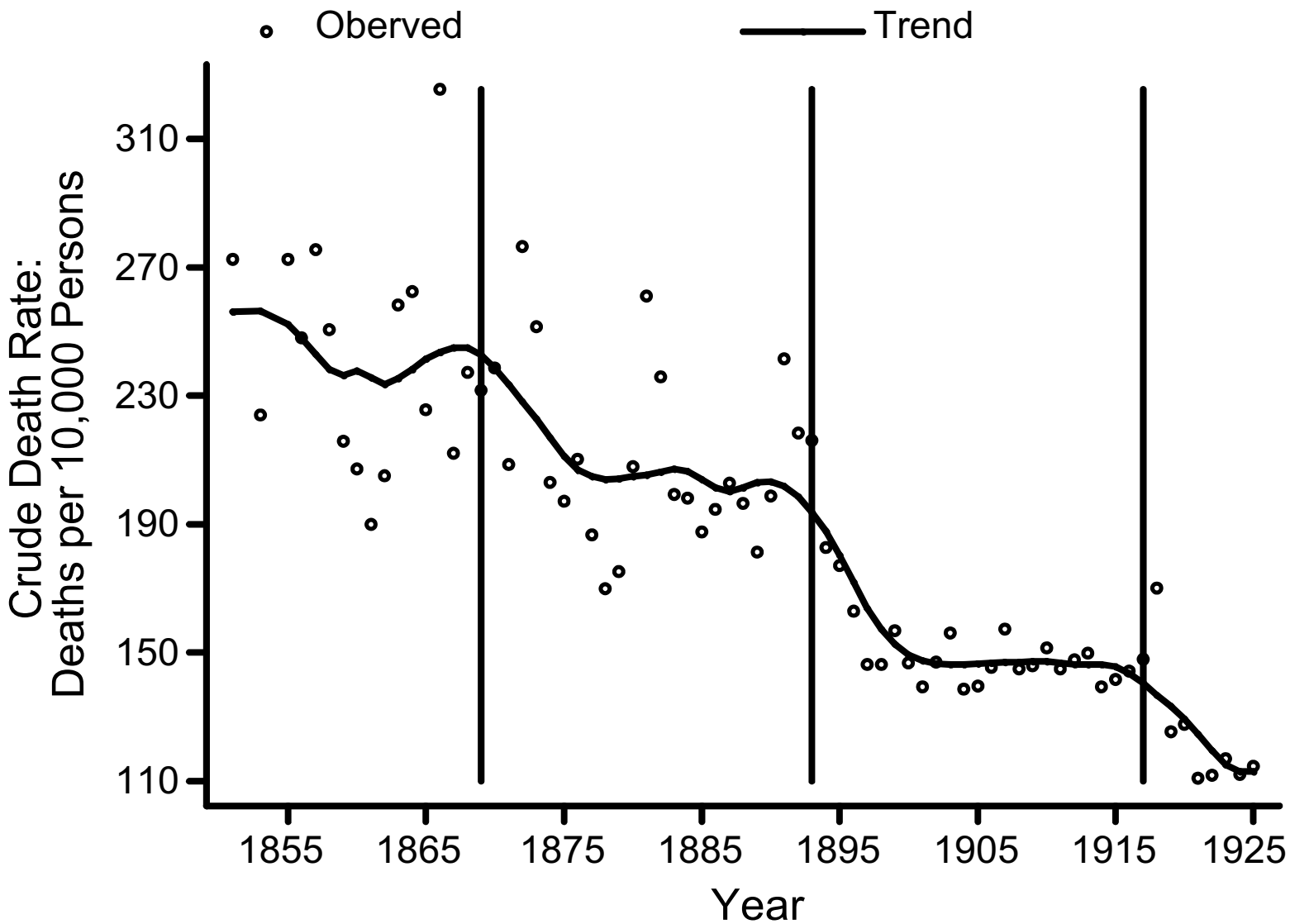
Ferrie and Troesken (2008), pp. 1–2, citing Haines (2001) for the other cities.

Epidemics in New York City, 1804–1930



Reported deaths per 1,000 people. Every labeled spike before 1900 is cholera, smallpox, or dysentery. After about 1890 the rate falls, with one large spike in 1918. Beach, Clay, and Saavedra (2022), *Journal of Economic Literature*.

The crude death rate in Chicago, 1853–1925



Deaths per 10,000 persons from all causes, with a lowess trend. The vertical lines mark the three changes in the city's water supply in 1867–71, 1893, and 1917. Three epidemic years with rates near 600 are left off the plot. Ferrie and Troesken (2008), Figure 2.

Four candidate explanations

- **Incomes and nutrition.** Industrialization raised wages and wages bought the housing and food that let bodies fight infection.
- **Medicine.** The diphtheria antitoxin arrived in the 1890s and antibiotics in the 1940s, after most of the gains in life expectancy.
- **Public investment.** Water intakes, filtration, sewers, and chlorination were the most expensive things a nineteenth-century city bought.
- **Cheap programs.** Vaccination, quarantine, and campaigns to change household behavior.

Ferrie and Troesken note that if incomes did the work, expensive waterworks "should take a back seat" to growth policy and the cheap programs become more attractive. Troesken explains why the cheap programs failed in America, Ferrie and Troesken measure what public investment bought, and Alsan and Wanamaker ask whether medicine gets used once it exists.

Ferrie and Troesken (2008), p. 2. Troesken (2015), pp. 10–11.

Disease control as a public good

Three inputs the state must supply

The claim

Controlling infectious disease requires three things from a state. **Coercion**, because vaccination and quarantine protect people other than the person compelled so voluntary take-up underprovides them. **Capital**, because water and sewer systems are lumpy, sunk, and financed by borrowing. And **trust**, because none of it works if citizens will not use what the state provides.

- Week 4's frame was state capacity and the constraints on it. Johnson and Koyama asked what lets a state tax, build, and enforce and what keeps it from expropriating. The same two questions govern public health.
- Troesken is about the constraints that limited coercion and enabled capital. Ferrie and Troesken measure what the capital bought. Alsan and Wanamaker measure what happens when the trust is destroyed.

The American constitutional order supplied the capital and hindered the coercion. At Tuskegee the federal government damaged the trust of older Black men creating a gradient in mistrust still present in 1998.

Commitment and the price of capital

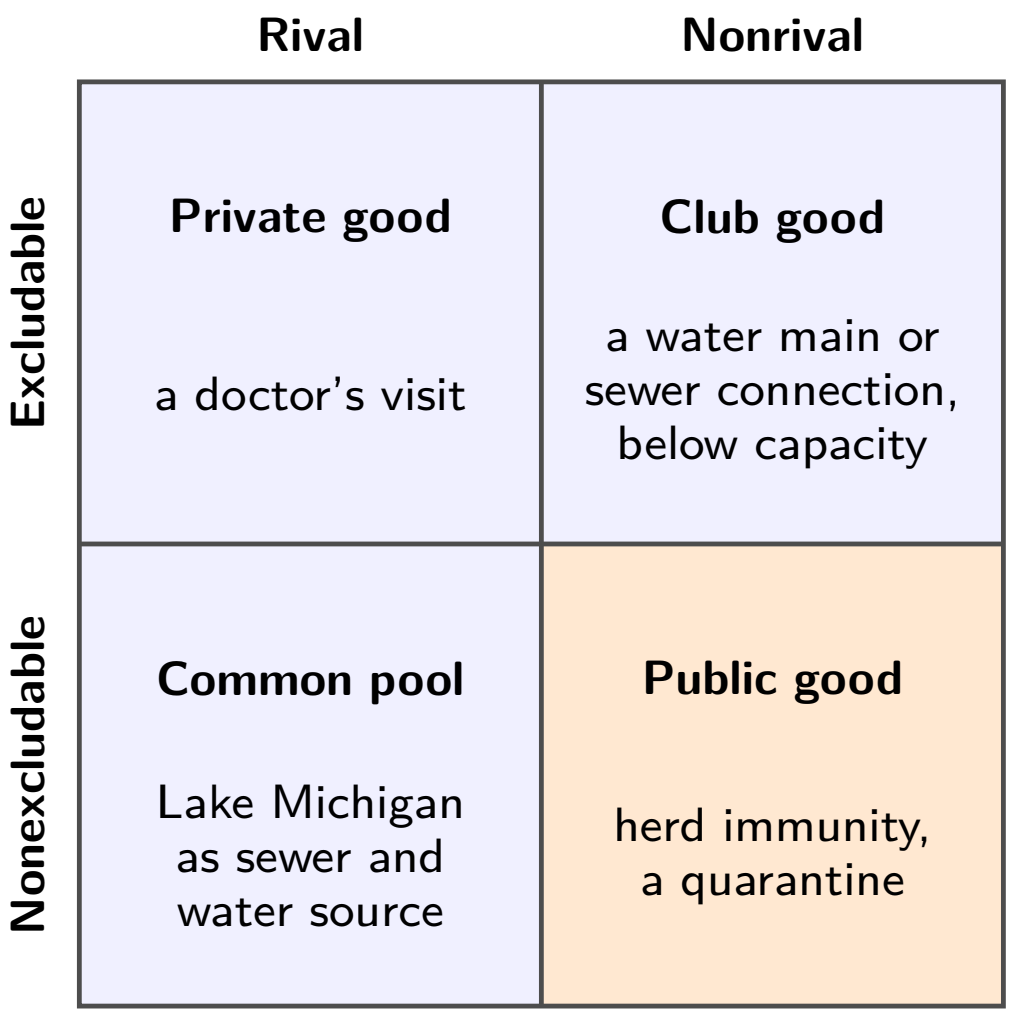
- A city that wants a waterworks has to borrow. A lender who expects the city council to repudiate the debt or tax the proceeds away will lend only at a price that makes the project unaffordable.

- Week 4 met this problem in North and Weingast's account of England after 1688. A sovereign that cannot renege borrows cheaply.

- The Contract Clause, Article I, Section 10, bars states from passing laws that impair the obligation of contracts. Troesken argues that the clause and state rules on municipal debt made it "much easier and cheaper for governments at all levels to raise the funds" (p. 8).

- In *Von Hoffman v. City of Quincy* (1867) Illinois had cut the tax that Quincy pledged to repay its bonds. The Supreme Court held that the cut impaired the bondholders' contracts and was void as applied to them.

Goods by rivalry and excludability



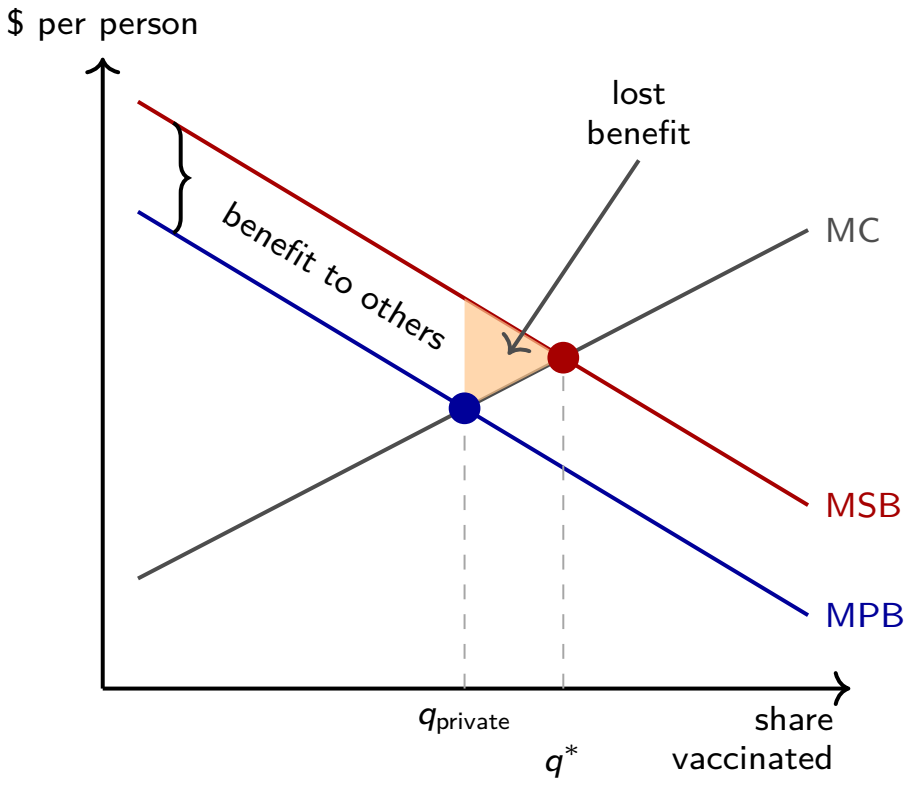
Rival. One person's use leaves less for others.

Excludable. A seller can keep out those who do not pay.

A smallpox vaccination is a private good with a public by-product. The person vaccinated is protected and so is everyone that person would otherwise have infected. Markets supply the private part and leave out the by-product.

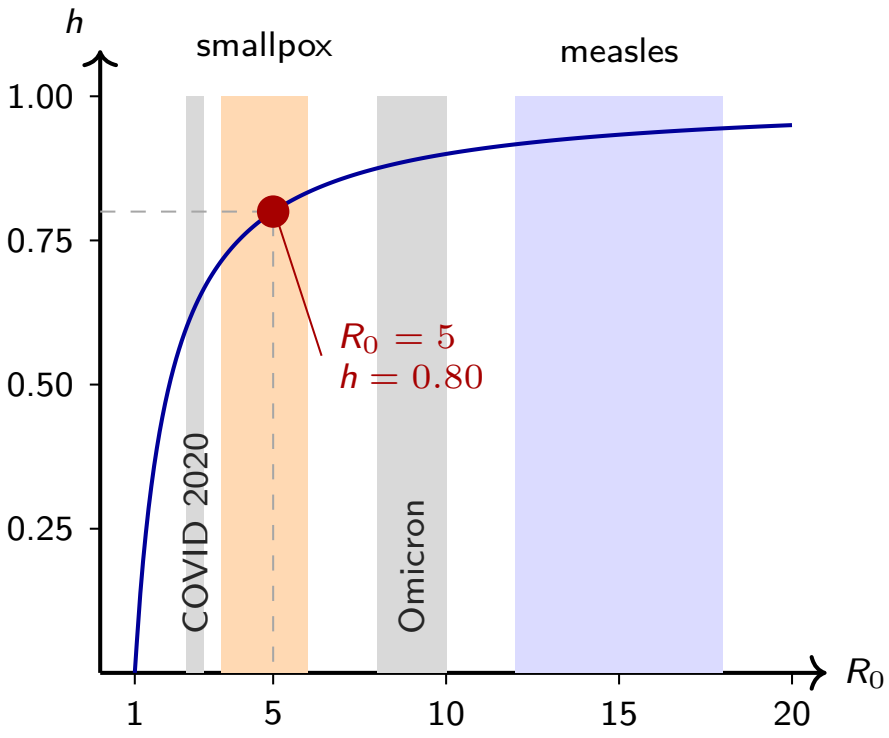
Chicago's lake was a common pool. Every sewer outfall lowered the quality of everyone's drinking water. The intake cribs were built to draw water beyond the outfalls.

Why voluntary vaccination falls short



- Each person compares a private benefit with a private cost, which includes real risks of the period's vaccine. Take-up stops at q_{private}.
- The protection a vaccinated person gives others is the gap between the private and the social benefit. Nobody is paid for it.
- Society would choose q^*. The shaded triangle is the benefit lost between the two.
- A subsidy or a mandate closes the gap. A mandate is the coercion input of the three-input frame.

Herd immunity



R_0 is the average number of people one case infects in a population with no immunity.

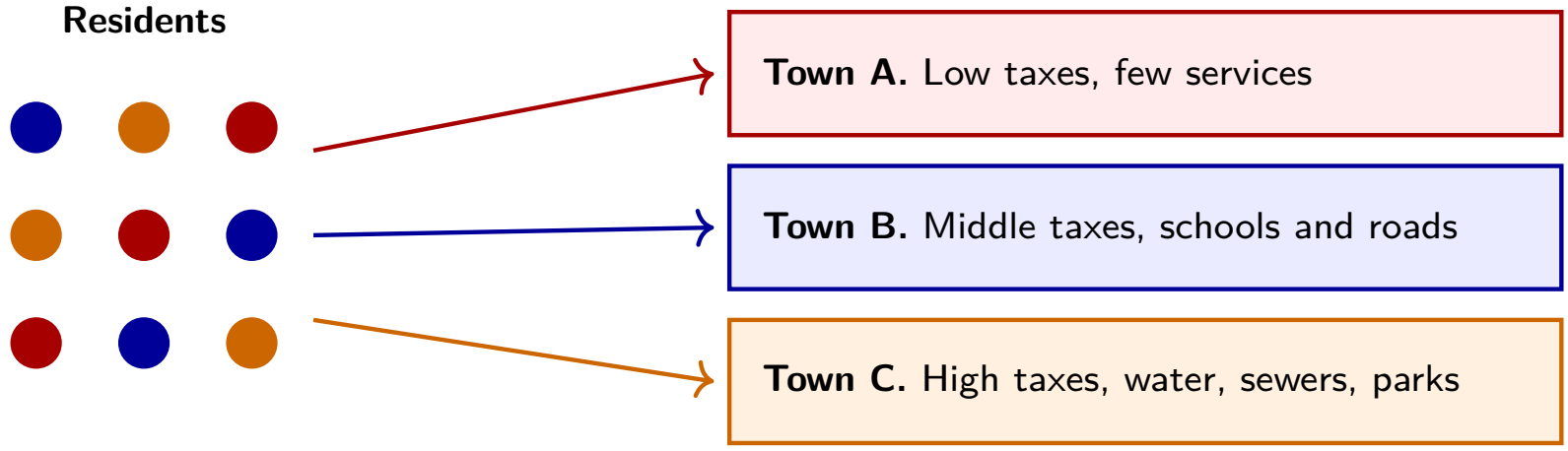
An outbreak dies out when each case infects fewer than one other person. That happens once a share h of the population is immune, where

$$h = 1 - \frac{1}{R_0}.$$

Below the herd-immunity threshold h the disease spreads. Above it chains of infection die out and people who are not immune are protected by everyone who is.

The formula assumes random mixing and a vaccine that works every time. With efficacy E the coverage needed is h/E. Smallpox R_0 of about 3.5–6 from Gani and Leach (2001). Measles is often cited at 12–18, though Guerra et al. (2017) find a wider range. COVID-19 about 2.5–3 for the 2020 strain (Liu et al. 2020) and 8–10 for Omicron (Liu and Rocklöv 2022).

Tiebout sorting



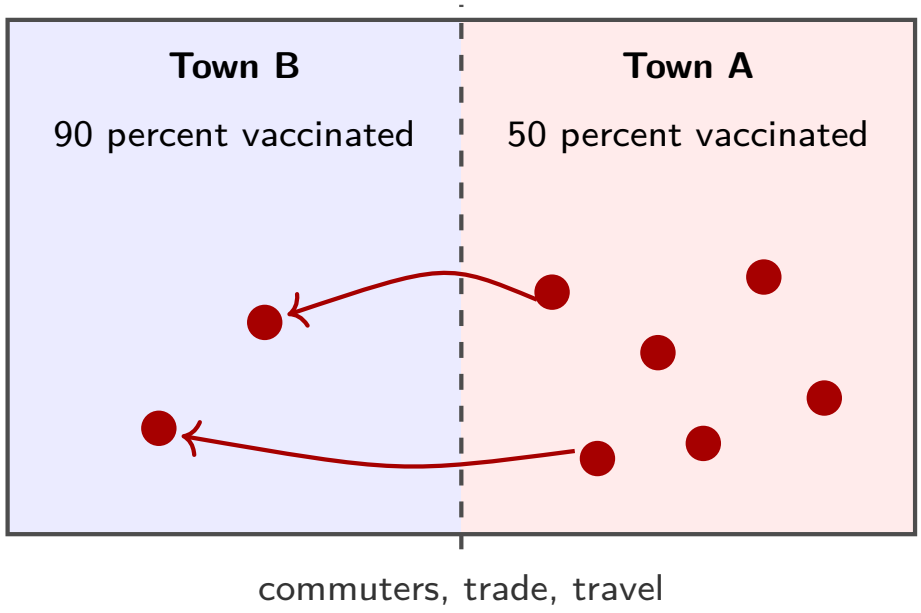
residents choose by moving

- Tiebout (1956). If people can move freely among many towns, each town offers a bundle of taxes and services and residents sort to the bundle they want. Residents' mobility reveals what they are willing to pay for.
- The result rests on assumptions, among them many jurisdictions, mobile residents, full information, and no spillovers between communities.

To the room. Which of those assumptions does an infectious disease break?

Tiebout (1956), pp. 418–420.

A spillover between two towns

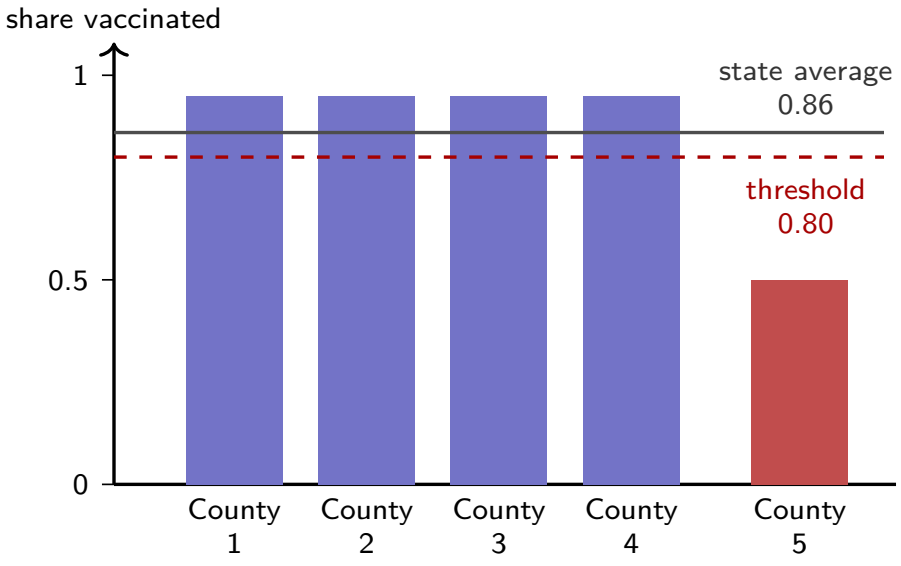


- A's voters decline to compel vaccination. The cases A's outbreaks cause in B do not enter their choice.
- B cannot vote in A. If A lies in another state, B's state may quarantine A's travelers. Nobody coordinates the states and courts limit measures that burden interstate commerce.
- Yellow fever worked the same way through ports. Troesken writes that at times the Commerce Clause "obstructed the development of a coherent and rational system of quarantines."

Tiebout sorting provides local public goods well when the benefits stay inside the town. Vaccination is a case where they do not.

Troesken (2015), Preface p. ix. The vaccination rates are illustrative.

An enclave can sit below the threshold inside a state above it



- Five counties of equal population. Four vaccinate at 95 percent and one, where objectors have gathered, at 50.
- The state average of 86 percent clears an 80 percent herd-immunity threshold. County 5 does not. Outbreaks start there and travelers carry cases into neighboring counties.
- Sorting makes this worse because objectors concentrate where the law lets them refuse.

Illustrative numbers, with the threshold for $R_0 = 5$. Troesken's North Dakota, whose smallpox death rate in the years smallpox circulated was about ten times that of states that compelled vaccination, is County 5 at the scale of a state.

Federalism's trade-off

What competition among governments did well

- Limited what governments could confiscate and made them compete for capital and labor, which is Weingast's (1995) case for market-preserving federalism.
- Gave rise to highly effective local sanitation programs, in Troesken's account. The borrowing that paid for them he credits to the Contract Clause.

What it did badly

- Coordinating against a disease that crosses borders, such as yellow fever arriving by ship.
- Holding vaccination rates above the threshold everywhere because objectors could sort into towns that would not compel them.

→ Congress gave the Marine Hospital Service quarantine powers in 1878 and 1893. Which level of government should have run vaccination in 1900 and what would it have cost?

Troesken (2015), Preface pp. viii–ix.

Troesken and the pox of liberty

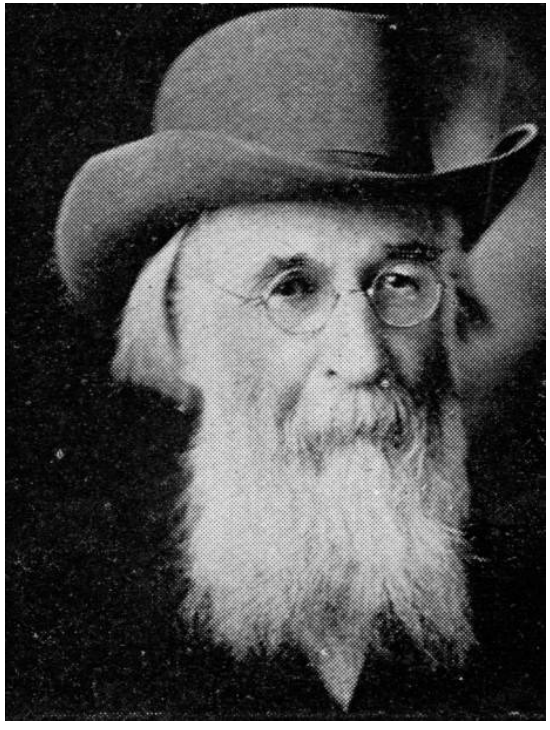
Smallpox

- **The disease.** The variola virus spread from person to person through breath droplets and contact with scabs, clothing, and bedding. About 12 days after infection came a high fever, then a rash that turned into deep pustules across the face and body.
- **The toll.** The common form, variola major, killed about 30 percent of those infected. Survivors were often scarred and sometimes blinded and were immune for life. A milder form, variola minor, killed about 1 percent and spread in the United States from the late 1890s.
- **Prevention.** Variolation, infecting a healthy person with matter from a smallpox sore, produced a milder case and was practiced in Boston by 1721. In 1796 Edward Jenner showed that cowpox protected at far lower risk. Protection faded over years, so revaccination mattered.
- **The end.** The last American case was in 1949 and the last natural case anywhere was in Somalia in 1977. The World Health Organization declared smallpox eradicated in 1980, the only human disease eradicated so far.

The descriptions are standard medical and historical summaries.

Justice Robinson of North Dakota

- James E. Robinson won election to the North Dakota Supreme Court in 1916 promising to "minister justice in a plain, common-sense, and businesslike manner." In *Rhea v. Board of Education* the Devils Lake school district had required proof of smallpox vaccination to enroll.
- The court ruled for the unvaccinated child. Robinson's concurring opinion, which he said rested on "the statutes, common knowledge, and pure reason," was a polemic.
- He believed that smallpox came from crowding and filth, that vaccination "slaughtered" 25,000 children a year, and that it caused "death, syphilis, cancer, consumption, eczema, [and] leprosy."



Justice James E. Robinson, c. 1919.

ND State Library, public domain.

North Dakota law barred excluding unvaccinated children from public school as late as 1975. In the decades when smallpox still circulated, the state's smallpox death rate was roughly ten times that of states that compelled vaccination.

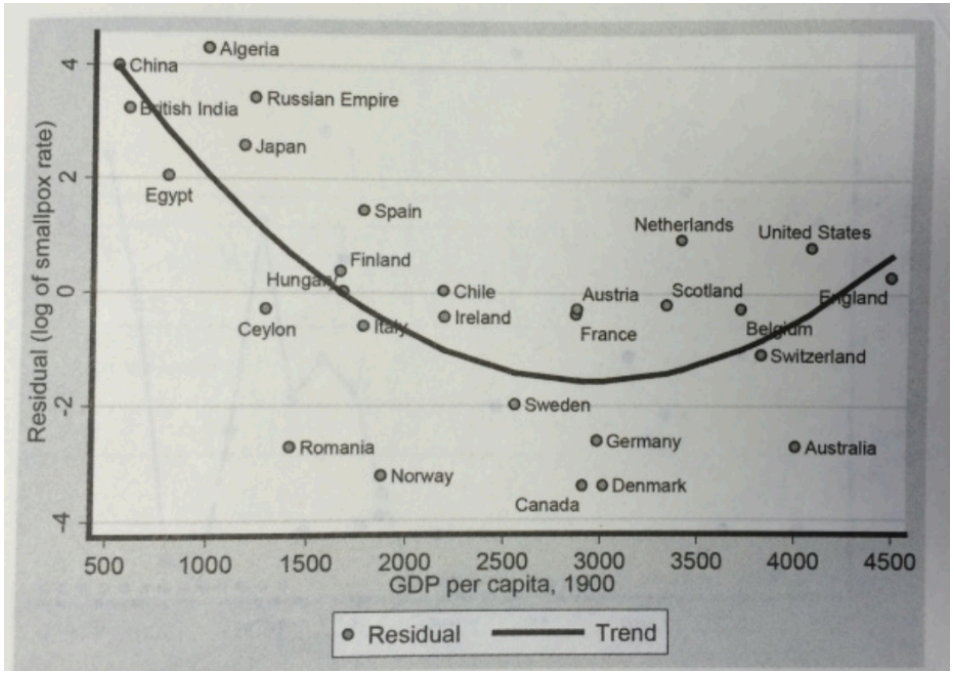
Troesken (2015), pp. 1–3. Robinson served 1917–1922. Troesken dates his election to 1918 and the court's records give 1916.

The puzzle

- Around 1900 the United States was the richest, most technologically advanced democracy in the world. It lagged poorer and less free societies in eradicating smallpox. Ceylon, now Sri Lanka and then a poor British colony, had a lower rate.

- The Soviet Union eradicated smallpox faster than the United States and England. Brazil's military coup of 1964 brought in a vaccination program far more intrusive than anything in the preceding century. In Troesken's account smallpox ended soon after.

- Troesken began with Sen's argument that democracies do not have famines because politicians answer to voters.



Smallpox rates against GDP per capita in 1900, after controlling for the Catholic share. Troesken (2015), Figure 4.6b, p. 99.

→ Why would Sen's logic fail for smallpox?

Troesken (2015), Preface pp. vii–viii and p. 3.

Troesken's answer to the puzzle

> Public-health law is as exposed to the tyranny of the majority as any other law. A minority's protection is a constitution enforced by courts. The protections that stop a majority from abusing a minority also let a dissenter refuse vaccination.



James Gillray, "The Cow-Pock, or the Wonderful Effects of the New Inoculation!", 1802, published by H. Humphrey and citing "the Publications of ye Anti-Vaccine Society." Library of Congress, public domain.

Troesken (2015), Preface pp. vii–viii. In Week 4 Madison's definition allowed a majority to be a faction.

Anti-vaccination as a libertarian movement

- *Mosher's Magazine*, 1902. The leaders of the movement were "the so-called intelligent ones," lawyers, writers, and teachers, people "most apt to deem themselves outside of the laws that make the ordinary human body sick or well."

- H. B. Anderson's tract called compulsory vaccination "medical slavery" and invoked the Thirteenth Amendment against it.

- Leicester, March 1885. Thousands marched under banners reading "Health and Liberty" and "The price of liberty is eternal vigilance."

- Troesken reports that British Egypt could not enforce vaccination on the English, whose smallpox rate ran six times the Egyptians', and that American troops enforcing vaccination after 1898 ended smallpox in Puerto Rico and Cuba within five years. New York and Massachusetts took another 50 years.

Troesken (2015), pp. 3–5. Troesken's reading of the comparisons is that the difference between places was consent.

The Commerce Clause and the Contract Clause

Two clauses that answer the faction problem of *Federalist* 10, which we read in Week 4, carry most of Troesken's argument.

	What it did for growth	**What it did for disease**
Commerce Clause	competition among jurisdictions for people and firms; credible commitment to investors (Weingast)	state quarantines against yellow fever stayed uncoordinated and open to challenge as burdens on commerce. Dissenters sorted into jurisdictions that would not compel vaccination
Contract Clause	no ex post alteration of debt contracts; credit markets revived after the Articles of Confederation	a market for municipal bonds. Cities could borrow to build water and sewer systems

Troesken (2015), Preface pp. viii–ix and pp. 6–9. Challenges to vaccination orders under the Fourteenth Amendment were often unsuccessful, but the litigation could delay policy. Where courts upheld vaccination, voters passed laws restricting health officials (p. 9).

Americans were unhealthy on some margins because they were rich and free

> "The language of modernity says we are unhealthy despite being rich and free. But the language of history suggests that we are unhealthy, on at least some margins, because we are rich and free and because our legal and political institutions function well."

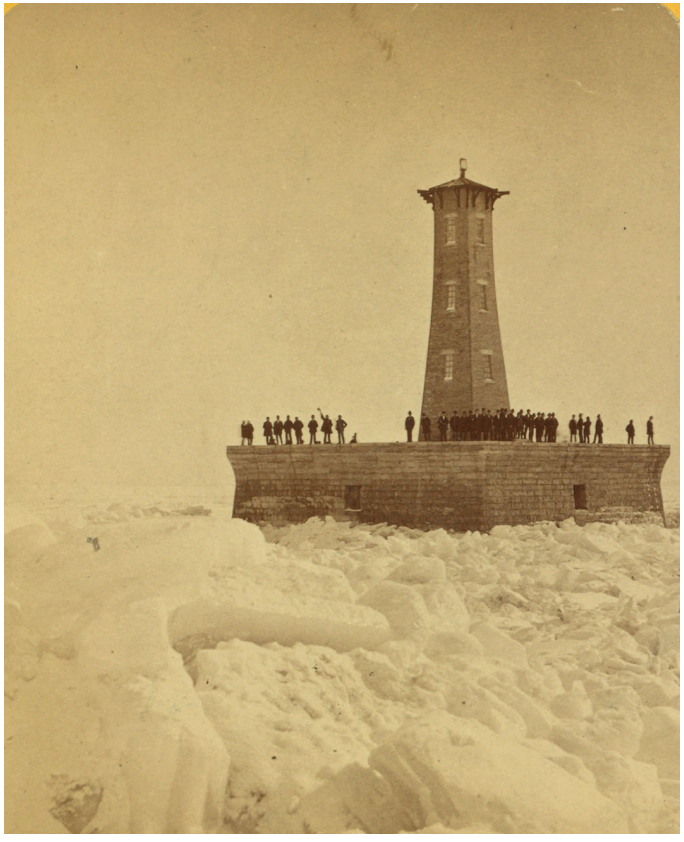
Troesken gives three reasons the trade-off matters (pp. 5–6).

1. The commitments to liberty that hindered smallpox control promoted economic growth and improved health on other margins.
2. Particular constitutional provisions aligned growth and health. The Contract Clause made possible the borrowing that financed waterworks, and the waterworks eradicated typhoid.
3. Public-health policy can oppress minorities. Health officials in the eugenics era sterilized people judged to have "socially undesirable" characteristics.

Troesken (2015), Preface pp. ix–x. He attributes about 60 percent of the mortality decline of the late nineteenth and early twentieth centuries to public water supplies and says his ideas "have not been formalized and subjected to statistical tests." Ferrie and Troesken's estimate for Chicago alone is lower, 35–56 percent of the decline from the 1870s to 1915–25.

Ferrie and Troesken on Chicago's water

The intake crib and the pumping station





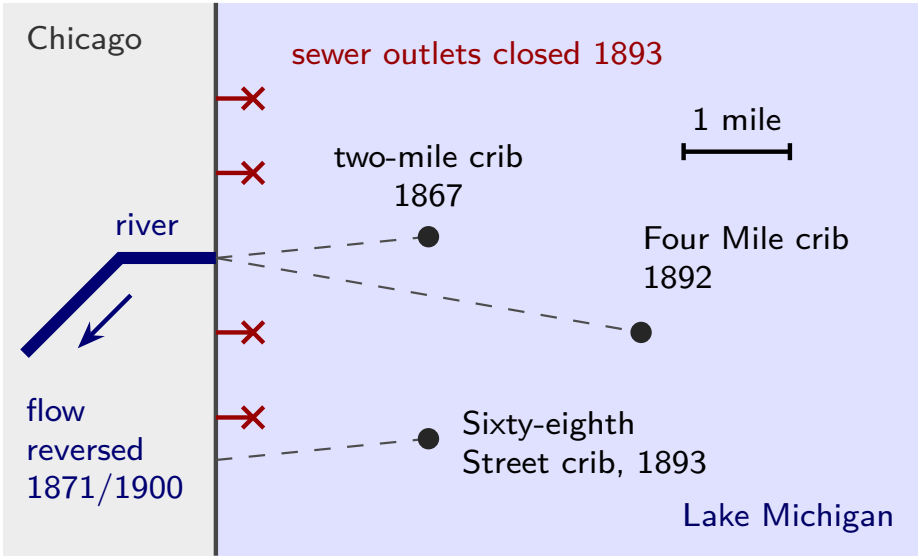
Left, the two-mile crib of 1867 in lake ice. Copelin & Son stereograph, New York Public Library, public domain. Right, the water tower and pumping station of 1869 from the lake side, 1893 or 1894. Ray Stannard Baker, Library of Congress, public domain.

Chicago's water supply in three regime changes

1867–1871. A two-mile tunnel under the lake to an intake crib away from the shore. The water tower and pumping station of 1869. The deepened Illinois and Michigan Canal of 1871, which reversed the Chicago River so that sewage ran away from the lake.

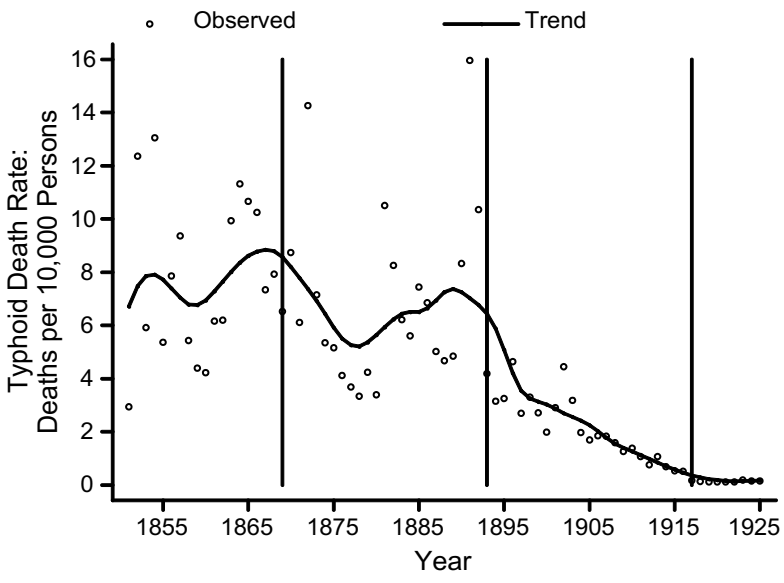
1893. The Four Mile crib (in operation by the end of 1892), the Sixty-eighth Street crib, and the permanent closure of every shoreline sewage outlet. The Sanitary and Ship Canal of 1900 held the reversal in storms.

1917. The Wilson Avenue crib and citywide chlorination.

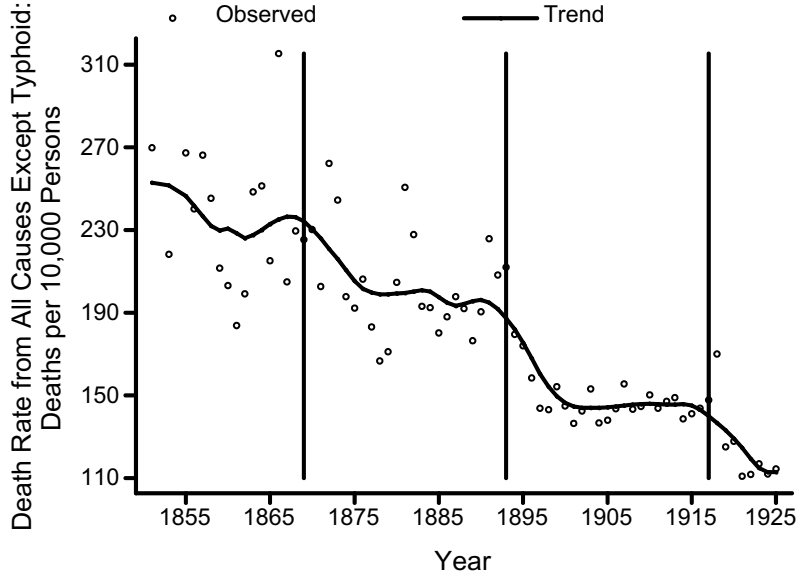


Ferrie and Troesken (2008), pp. 2–3, from Cain (1977). The canal follows the Chicago portage of Week 6. In Week 2 the Lambeth company of Snow's study moved its intake upstream of London's sewage in 1852. Chicago moved its intake offshore. Crib positions are schematic.

Typhoid and everything else fell at the same dates



Typhoid



All causes except typhoid

- Typhoid averaged 7.3 deaths per 10,000 before 1893 and never passed 5 after.
- The total death rate fell from 200–250 to about 110 (their Figure 2).

To the room. Water carries typhoid. What could produce the right-hand panel?

Ferrie and Troesken (2008), Figures 1 and 3, p. 3.

Three explanations for the second panel

1. **Coincidence.** Other public-health measures arrived at the same time. Pasteurization was not mandatory in Chicago until the early 1900s and there is little evidence it had any appreciable effect. The diphtheria antitoxin of the 1890s worked, but diphtheria was only 5 percent of deaths.

2. **Behavior.** Households changed their habits at the same moment. Then why that moment and why did officials across the United States and Europe record the same pattern after filtration plants opened?

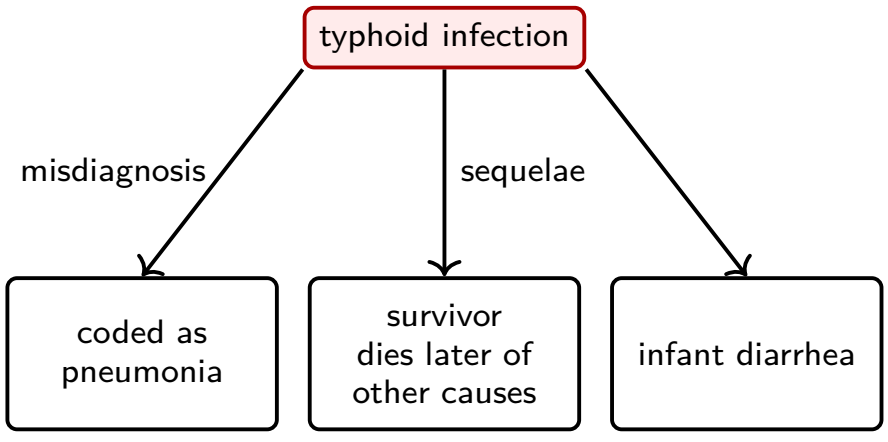
3. **Diffuse effects of clean water.** Typhoid killed few of its victims directly and damaged all of them. Removing typhoid removed deaths that were coded to other causes.

Ferrie and Troesken (2008), p. 4. The paper tests the third. In March 1890 the *Chicago Tribune* priced sewage in the water at $200,000 a month and asked the council to act. The 1891–92 epidemic, the spike in the typhoid panel on the previous slide, followed. The authors add that the *Tribune* overstated the city's inaction, since its earlier works had been overwhelmed by population growth (pp. 14–15, n. 15).

The Mills–Reincke phenomenon

Definition

For every death from typhoid fever that water purification prevents, several deaths from causes not usually thought of as waterborne are also prevented.



each is counted under another cause

- Mills in Lawrence, Massachusetts, and Reincke in Hamburg saw it independently around 1893. Hazen and Sedgwick's average was two to four other deaths per typhoid death.
- Case fatality was 5–10 percent. The rash that confirmed typhoid appeared in 5–20 percent of cases and the Widal test came only in the late 1890s.
- Union Army recruits who had typhoid carried about a 25 percent higher risk of heart trouble later (Costa 2000).

Sequelae are lasting conditions an illness leaves behind, such as the heart and kidney damage typhoid left in survivors. Dublin (1915) followed 1,574 typhoid survivors for three years. They died at three times the rate of others in the first year and twice the rate in the second. Of those who died, 39 percent died of tuberculosis and 23 percent of heart failure.

Ferrie and Troesken (2008), pp. 5–8.

Ferrie and Troesken's specification

A time series for the city from the annual reports of the Chicago Department of Health, in deaths per 10,000.

$$y_t \;=\; \beta_0 + \beta_1\, x_{t-k} + \beta_2\, y_{t-1} + \beta_3\, \text{Year}_t + \varepsilon_t, \qquad k \in \{0, 1, 2\}$$

Here y_t is the death rate from causes other than typhoid and x_t is the typhoid death rate. Four epidemic-year dummies enter. Newey–West standard errors allow the errors to be correlated across up to three years.

The ward panel uses the city's 34 wards for 1891–1894 and 1896–1900, $y_{it} = \alpha_i + \tau_t + \beta_1 x_{it} + \varepsilon_{it}$, with ward and year effects.

> This regresses one death rate on another. β_1 is the number of non-typhoid deaths that move with one typhoid death, holding a trend and last year's rate fixed. It identifies the Mills–Reincke factor only if nothing else that kills people moves with typhoid in the same year. The year effects in the panel remove shocks common to all of Chicago's wards.

Ferrie and Troesken (2008), pp. 9–10, equations (1) and (2).

Deaths from other causes per typhoid death

	1855–1925		Before 1880		After 1867	
	(1)	(2)	(3)	(4)	(5)	(6)
Typhoid$_t$	9.90		15.8		4.37	
	(4.39)		(6.42)		(.905)	
Typhoid$_{t-1}$	1.65	4.34		7.30		4.00
	(.435)	(1.12)		(2.15)		(1.58)
N	76	76	28	28	58	58

- Over 1855–1925 one typhoid death went with 9.9 deaths from other causes in the same year (column 1). Entered alone in column (2), last year's typhoid rate went with 4.3.

- Before 1880 the same-year number is 15.8, against 4.4 after 1867. The authors read the gap as consistent with misdiagnosis. The samples overlap, neither isolates the Widal test of the late 1890s, and the pre-1880 estimate is imprecise.

Ferrie and Troesken (2008), Table 1, with the period headings as printed. The dependent variable is the death rate from all causes except typhoid. All columns include a lagged dependent variable, a trend, and epidemic-year dummies.

Deaths by cause and the other-disease placebo

Table 2, by cause

- Infants under one, 1.65 extra deaths per typhoid death. Children one to four, about 1.0.
- Kidney failure, 0.8.
- Tuberculosis and pneumonia together, 1–1.5.
- Heart failure, no effect.

Table 3, other diseases as the regressor

- Pneumonia, bronchitis, whooping cough, and diphtheria have no effect on deaths from other causes.
- Influenza and scarlet fever have a negative effect. They killed people who would soon have died of something else.
- Tuberculosis is 2.0 and significant. Measles is 2.5 and insignificant. Each is about half of typhoid's post-1867 coefficient of 4.4. With the lag the tuberculosis effect turns negative and measles holds up, still well below typhoid.

Table 3 checks that the method does not find a factor for every disease. If pneumonia also "caused" five other deaths the typhoid coefficient would be picking up a common shock.

Ferrie and Troesken (2008), Tables 2 and 3, pp. 11–12. The kidney figure is from the text, since printed Table 2's kidney columns repeat the child columns.

The ward panel

To the room. If shocks common to the whole city drove the time-series result, what should adding year effects do?

	(1)	(2)	(3)	(4)	(5)
Non-typhoid$_{t-1}$				.884	.946
Typhoid$_t$	5.22	5.00		2.32	
Typhoid$_{t-1}$			3.90		.015
Ward effects	yes	yes	yes	yes	yes
Year effects	no	yes	yes	yes	yes
N	306	306	306	272	272

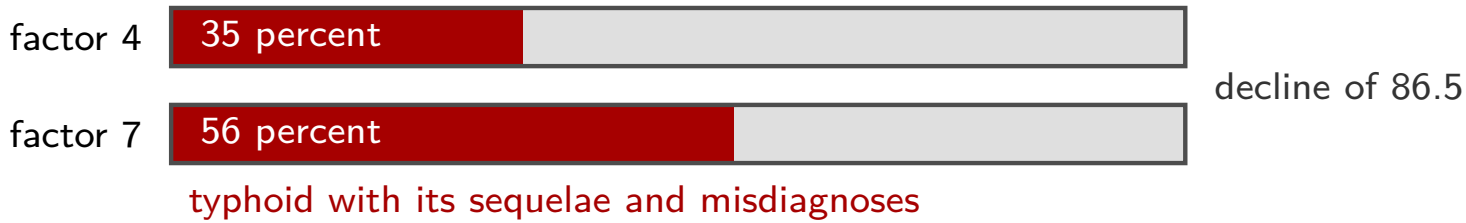
- Year effects move the coefficient from 5.2 to 5.0, a fall of 4 percent.
- Adding last year's ward death rate cuts it to 2.3. With that lag at 0.88 the 2.3 is a one-year effect and the implied long-run effect, $2.3/(1-0.88)$, is near 20.
- With the lagged typhoid rate the coefficient is zero, which the authors call "the only piece of evidence we can find that is inconsistent with the Mills–Reincke phenomenon."

Ferrie and Troesken (2008), Table 4, 34 wards and nine years. In a panel this short the lag coefficient is biased downward (Nickell 1981) and the bias passed to the typhoid coefficient has no general sign.

The accounting

The factor is the Mills–Reincke factor, the number of deaths from other causes that came with each typhoid death. The paper's estimates usually fall between 4 and 7 and the authors use those two values (p. 13).

Deaths per 10,000	1870–79	1915–25	Change
All causes	211.8	125.3	86.5
Typhoid	6.2	0.2	6.0



- Typhoid fell by 6.0. At a factor of 4 each of those deaths carried four more, so water's share of the decline is $6.0 \times 5 = 30$ of 86.5, or 35 percent. At a factor of 7 it is $6.0 \times 8 = 48$, or 56 percent. The abstract says 30–50 percent.
- The same arithmetic on the 1870s level puts typhoid and its sequelae and misdiagnoses at 15–23 percent of all deaths, against 3 percent for typhoid alone.

Ferrie and Troesken (2008), Table 5 and p. 13, excluding 1918. At the pre-1880 factor of 15.8 the same arithmetic credits typhoid with more than the whole decline.

The return on the waterworks

Factor	Lives saved a year	Cost per life	Annual return
0	1,621	$650,636	40 percent
4	8,105	$130,127	202 percent
7	12,968	$81,330	324 percent

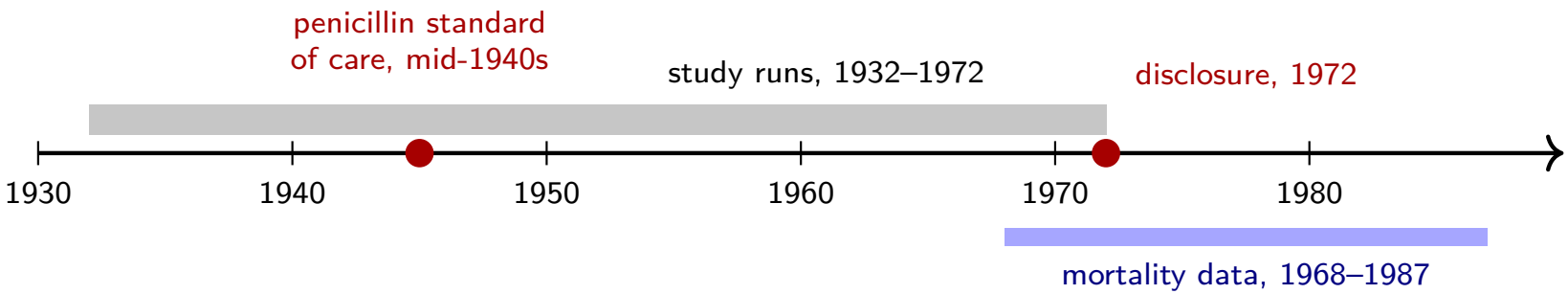
- The Census valued Chicago's waterworks at $52.6 million in 1915, about $1.05 billion in 2005 dollars. A life is valued at 30 years of a manufacturing wage discounted at 5 percent, about $263,000 in 2005 dollars.

Ferrie and Troesken (2008), Table 6, pp. 14–15.

Alsan and Wanamaker on Tuskegee

The Tuskegee study, 1932–1972

- From 1932 to 1972 the United States Public Health Service ran the "Tuskegee Study of Untreated Syphilis in the Negro Male." It followed about 600 Black men in Macon County, Alabama, about two-thirds of whom had syphilis, to observe the natural course of the disease.
- Survivors later reported that study doctors diagnosed them with "bad blood" and most believed they were being treated. They received physical exams, hot meals, and burial stipends and were subjected to blood draws, spinal taps, and eventually autopsies.
- Penicillin became the standard of care by the mid-1940s. It was withheld and the men were discouraged from seeking care elsewhere.
- Jean Heller of the Associated Press reported the study in 1972. It was halted that year. Most of the subjects were dead and dozens of wives and children had been infected.



Alsan and Wanamaker (2018), pp. 408–409 and 414.

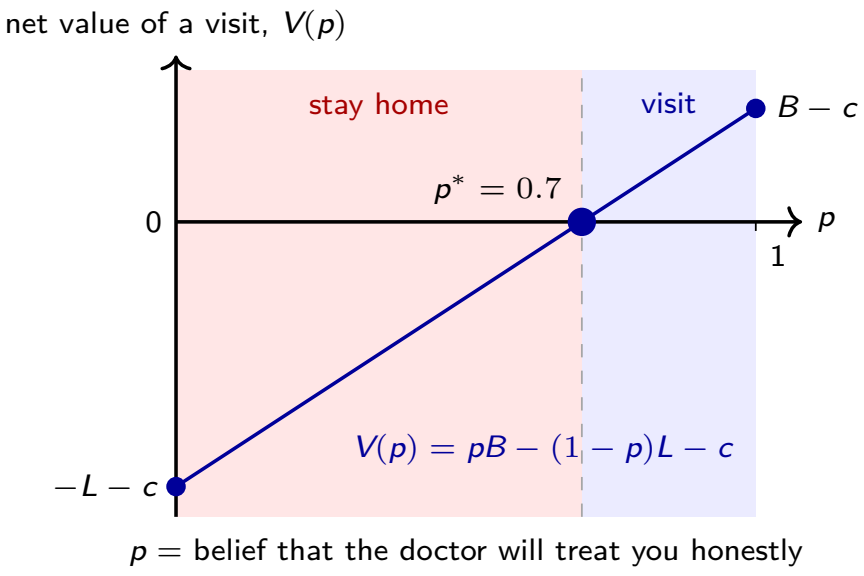
The question

- In recent life tables a Black man of 45 can expect 3 fewer years of life than a white man of 45 and 5 fewer than a Black woman. Income, education, and insurance explain part of the gap and leave a residual.
- Qualitative work blames mistrust of the medical system. Does mistrust cause the disparity, and how much of it?
- Every prior study of Tuskegee's legacy fails on at least one of four counts. All were run 20 or more years after 1972. A correlation between mistrust and knowing about Tuskegee cannot show which causes which. The outcome was willingness to join a clinical trial. The samples were local.

The disclosure is a shock to beliefs with a date, a target group, and a place. The prior literature lacked all three.

Alsan and Wanamaker (2018), pp. 408 and 415.

Medical care as a credence good

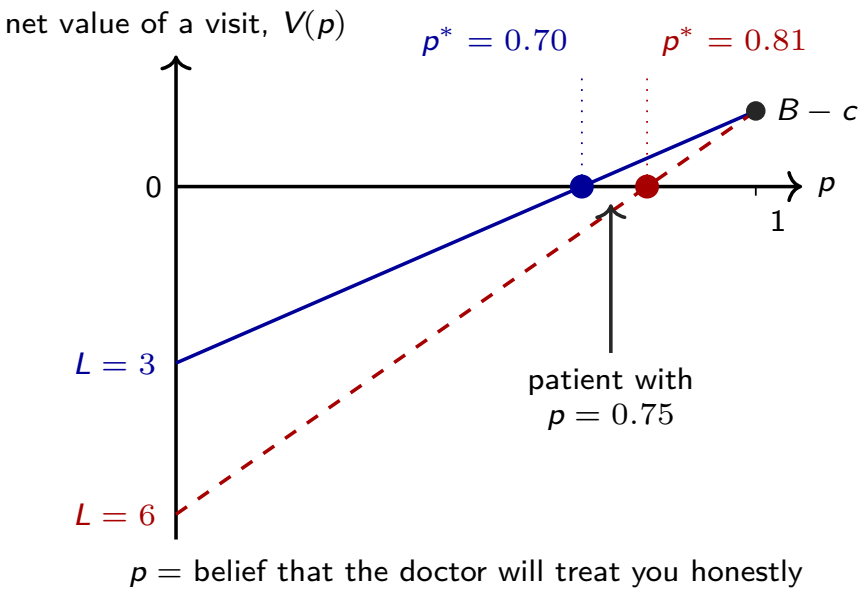


- A **credence good** is one whose quality the buyer cannot judge in normal use, even afterward. Most patients cannot tell whether the treatment they received was the right one.
- So the decision rests on a belief p that the doctor will treat you honestly. Honest care is worth B, dishonest care costs you L, and the visit costs c.
- Visit if $pB - (1-p)L > c$, that is, if p is above

$$p^* = \frac{c+L}{B+L}.$$

Illustrative values $B = 2$, $L = 3$, and $c = 0.5$ give $p^* = 0.7$.

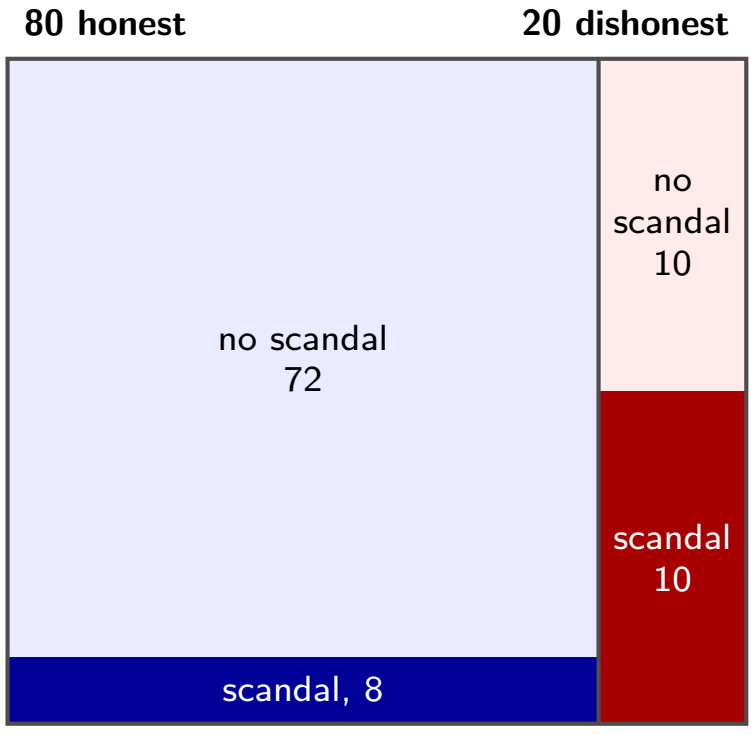
A larger expected loss raises the threshold



- Suppose the news raises the harm a patient expects from dishonest care, from $L = 3$ to $L = 6$. The line pivots down around $B - c$, the value of a visit to an honest doctor.
- The threshold rises from 0.70 to 0.81. A patient with a belief of 0.75 visited before the news and stays home after it, though the belief has not changed.
- In general $\partial p^*/\partial L = (B - c)/(B + L)^2$, which is positive whenever an honest visit is worth its cost.

Illustrative values, with $B = 2$ and $c = 0.5$ held fixed. The news can work through L, as here, or through p, as on the next slides. Both reduce visits.

Bayes' rule in natural frequencies



Take 100 medical systems like yours. You think 80 are honest. A scandal like Tuskegee comes to light in 10 percent of honest systems and in 50 percent of dishonest ones.

You hear of a scandal. The systems with one are the dark cells, 8 honest and 10 not, so

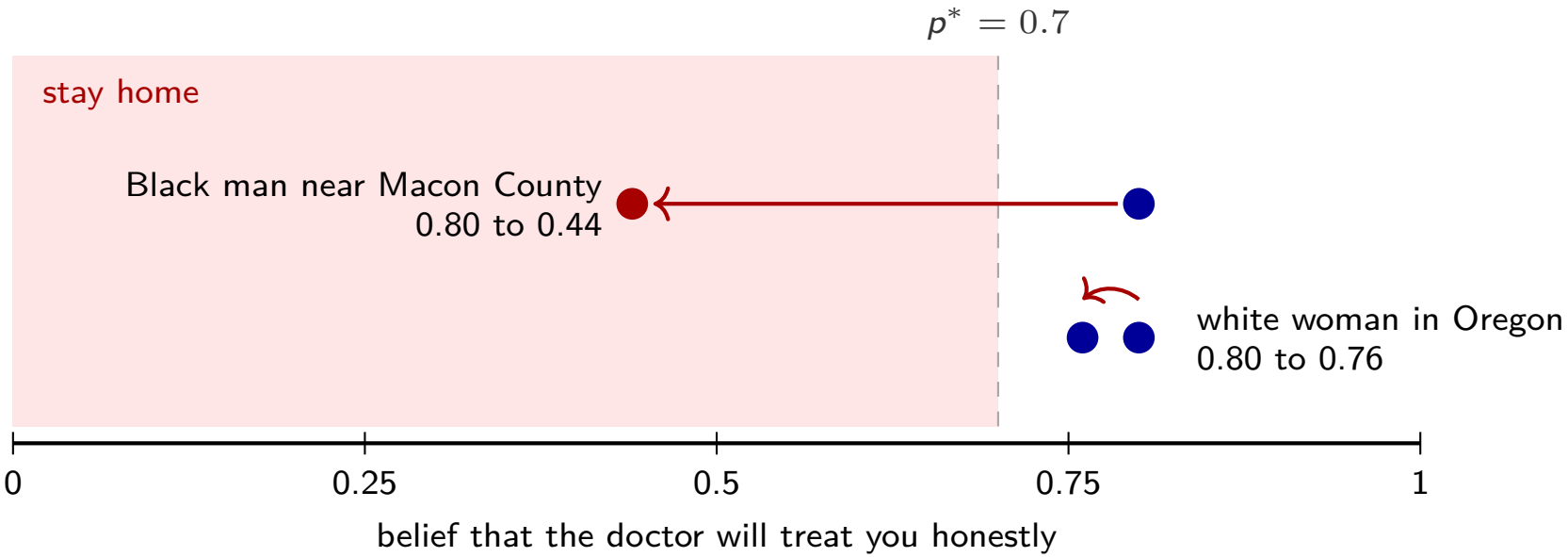
$$\Pr(\text{honest} \mid \text{scandal}) = \frac{8}{8+10} \approx 0.44.$$

That is Bayes' rule,

$$\Pr(H \mid s) = \frac{\Pr(s \mid H)\Pr(H)}{\Pr(s \mid H)\Pr(H) + \Pr(s \mid D)\Pr(D)}.$$

Illustrative numbers. The belief falls from 0.80 to 0.44, below the visit threshold of 0.7 on the previous slide.

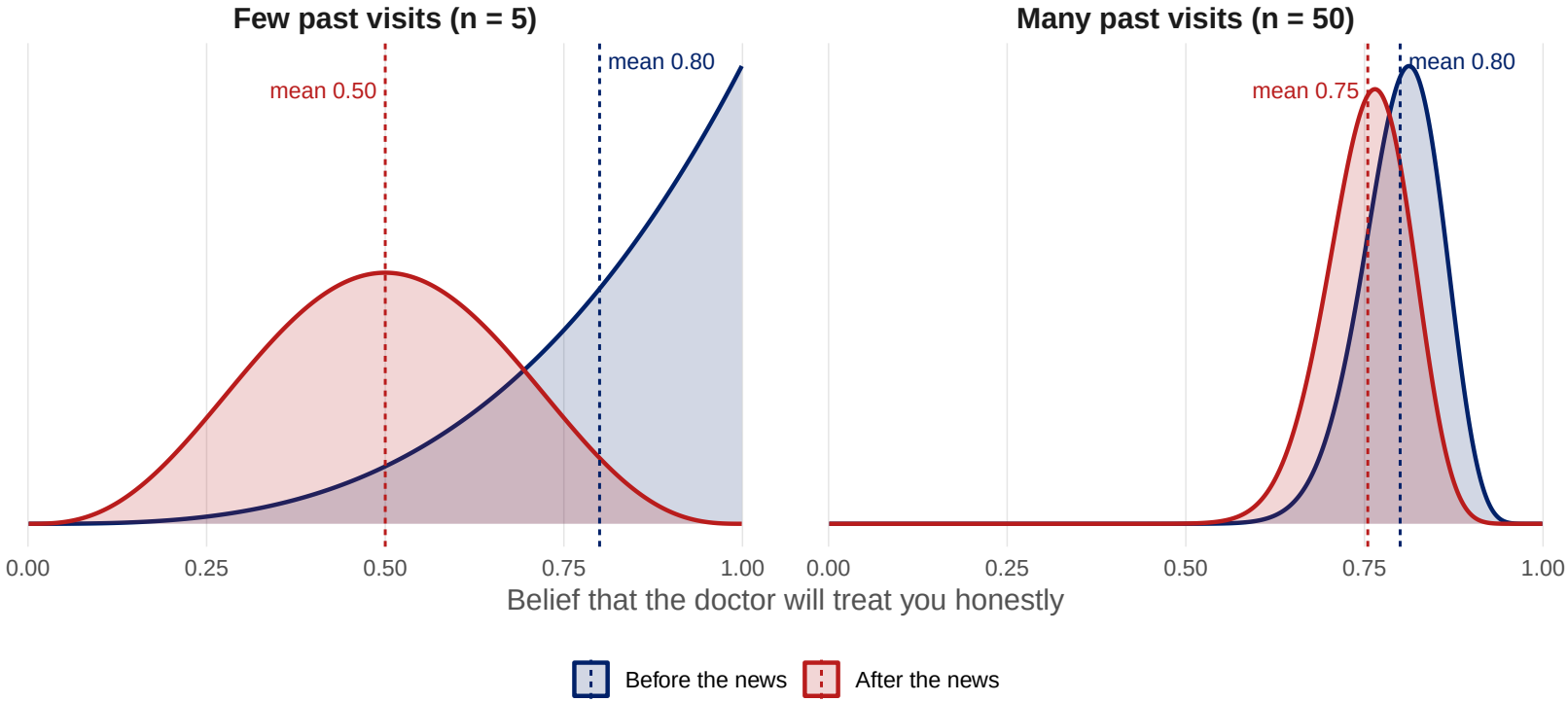
Two readers of the same news



- The first reader sees the scandal as evidence about a system like the one he uses. For him a scandal is five times as likely in a dishonest system as in an honest one.
- The second sees it as news about someone else. For her it is nearly uninformative, a likelihood ratio of 1.25 against 5, so her belief barely moves and stays above the visit threshold.
- The ratio $\Pr(s \mid D)/\Pr(s \mid H)$ measures how informative the news is to the reader. Identification with the victims raises it.

Illustrative numbers. Tabellini (2008) models norms of cooperation that weaken with distance between the parties, measured in any space. Here the relevant distances are race, sex, class, and place.

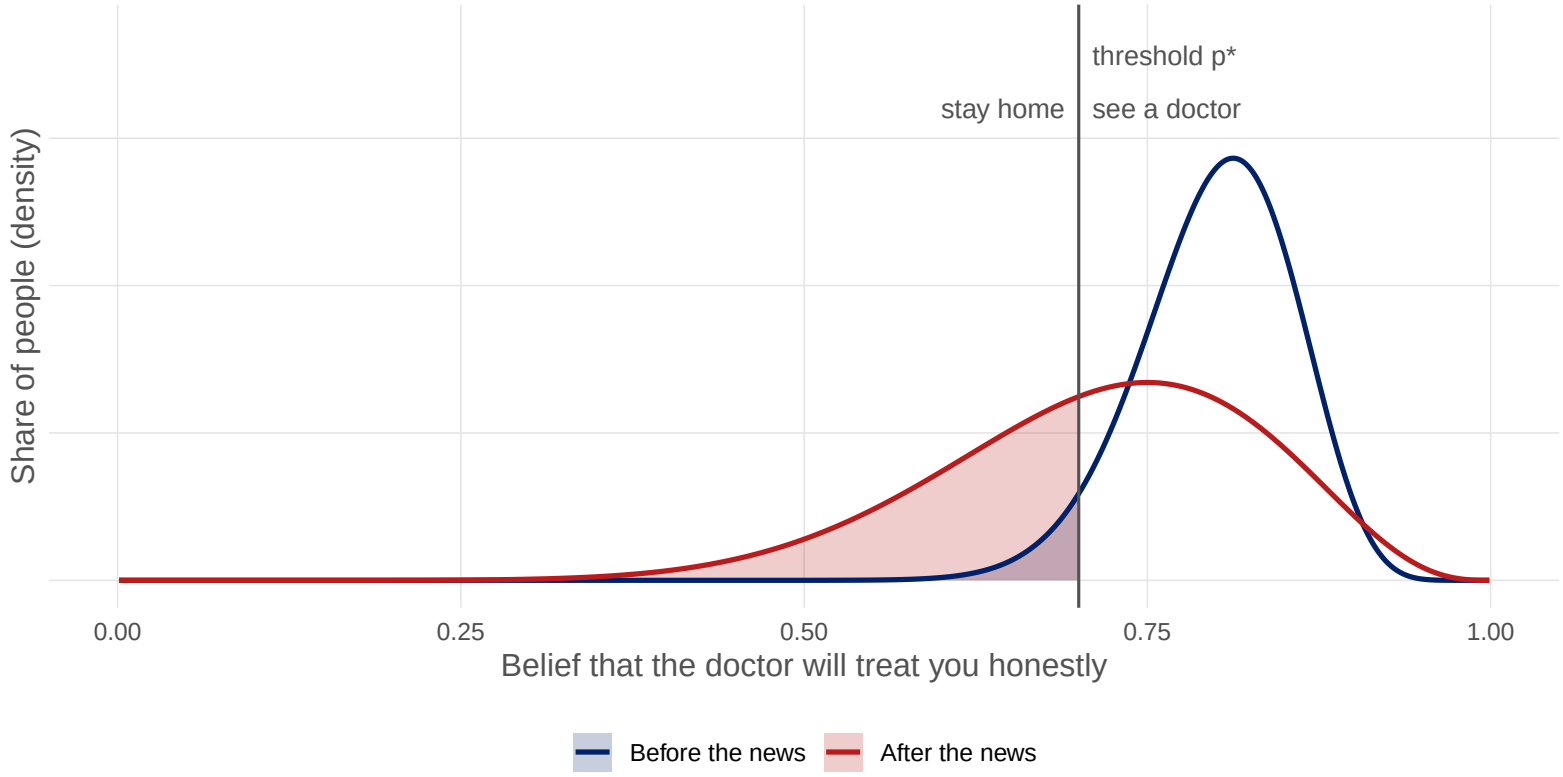
Experience makes a belief harder to move



- To let experience matter, treat the belief as an estimate built from past visits. Both people start at 0.80, one with five visits behind the belief and one with 50, and the news counts as three bad visits.
- The posterior weights past experience by n and the news by w. With bad visits scored zero, $p_1 = np_0/(n + w)$. With $p_0 = 0.8$ and $w = 3$ it is 0.50 after five visits, below the visit threshold, and 0.75 after 50, above it.

Illustrative Beta priors, from `scripts/R/wk08_bayes_figures.R`. Each curve is one person's uncertainty. Veterans and mothers have many past visits. Alsan and Wanamaker's multiperiod model has the same feature (p. 448).

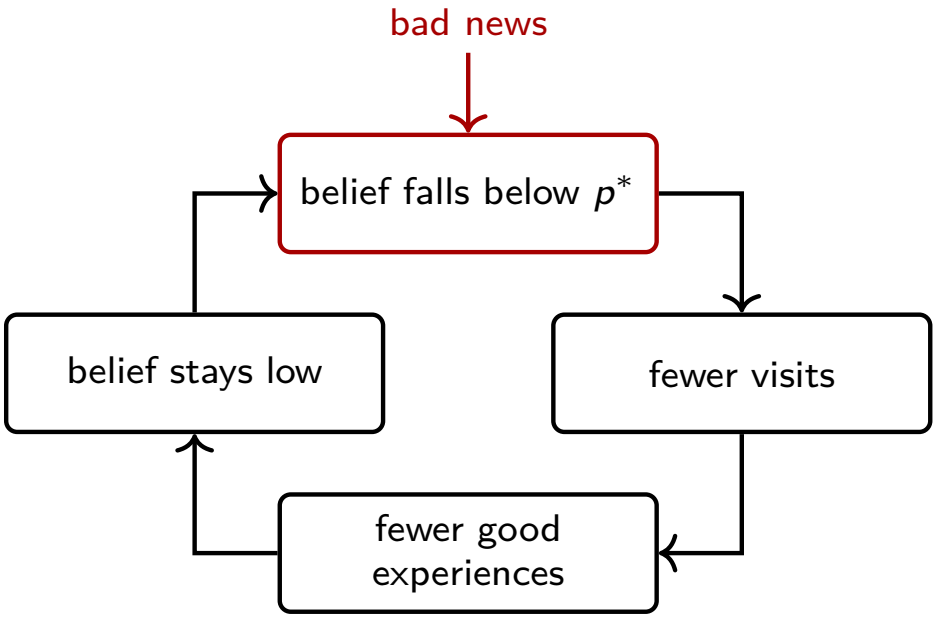
From beliefs to visits



- Each curve spreads point beliefs across people and the after-news curve is assumed. The average falls only from 0.80 to 0.71, yet the share below the threshold rises from about 5 percent to 42.
- In Alsan and Wanamaker most of the response is fewer visits by men who still go. The chance of any visit falls 4 percent and the number of visits 22 percent. Treat the threshold as a stylization.

Illustrative Beta distributions with $p^* = 0.7$. Alsan and Wanamaker measure the visits and the 1998 General Social Survey measures the beliefs.

Persistence of the belief



- Good experiences are what would raise the belief again and they come only from visits.
- A person who stops going stops collecting the evidence that could change her mind. Denrell and March (2001) call this the hot stove effect.
- Alsan and Wanamaker find a gradient in mistrust of doctors in 1998, 26 years after the disclosure, though no earlier measure exists.

Nunn and Wantchekon (2011) found that Africans whose ancestors were more exposed to the slave trade trust less today. Alsan and Wanamaker find no such gradient among respondents younger than ten in 1972 (n. 36), which fits beliefs that persist within a lifetime.

The hot stove effect is one explanation of the persistence. Alsan and Wanamaker do not test it.

What the belief frame predicts

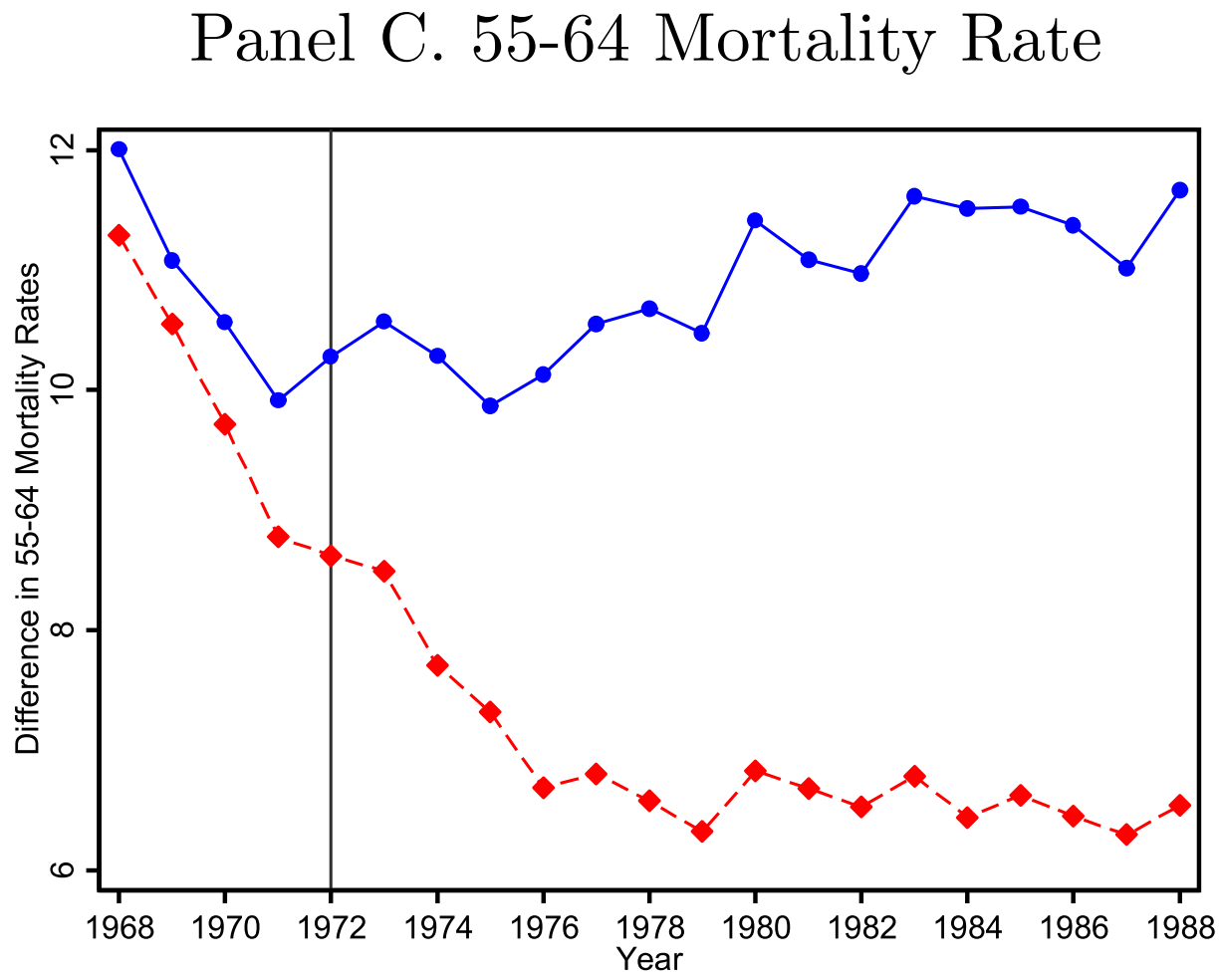
- **Black men** should respond more than Black women and more than white men because the victims were Black men. Men's more elastic demand for care predicts the gap between Black men and Black women without any belief.

- **The poor and less educated** should respond more because the victims were poor and largely illiterate.

- **Nonveterans** should respond more than veterans, and women without children more than mothers, because veterans and mothers have a longer history with the medical system.

- **Proximity** should matter if the news is about the local medical system or if identification falls with distance.

Older Black men stopped converging after 1972



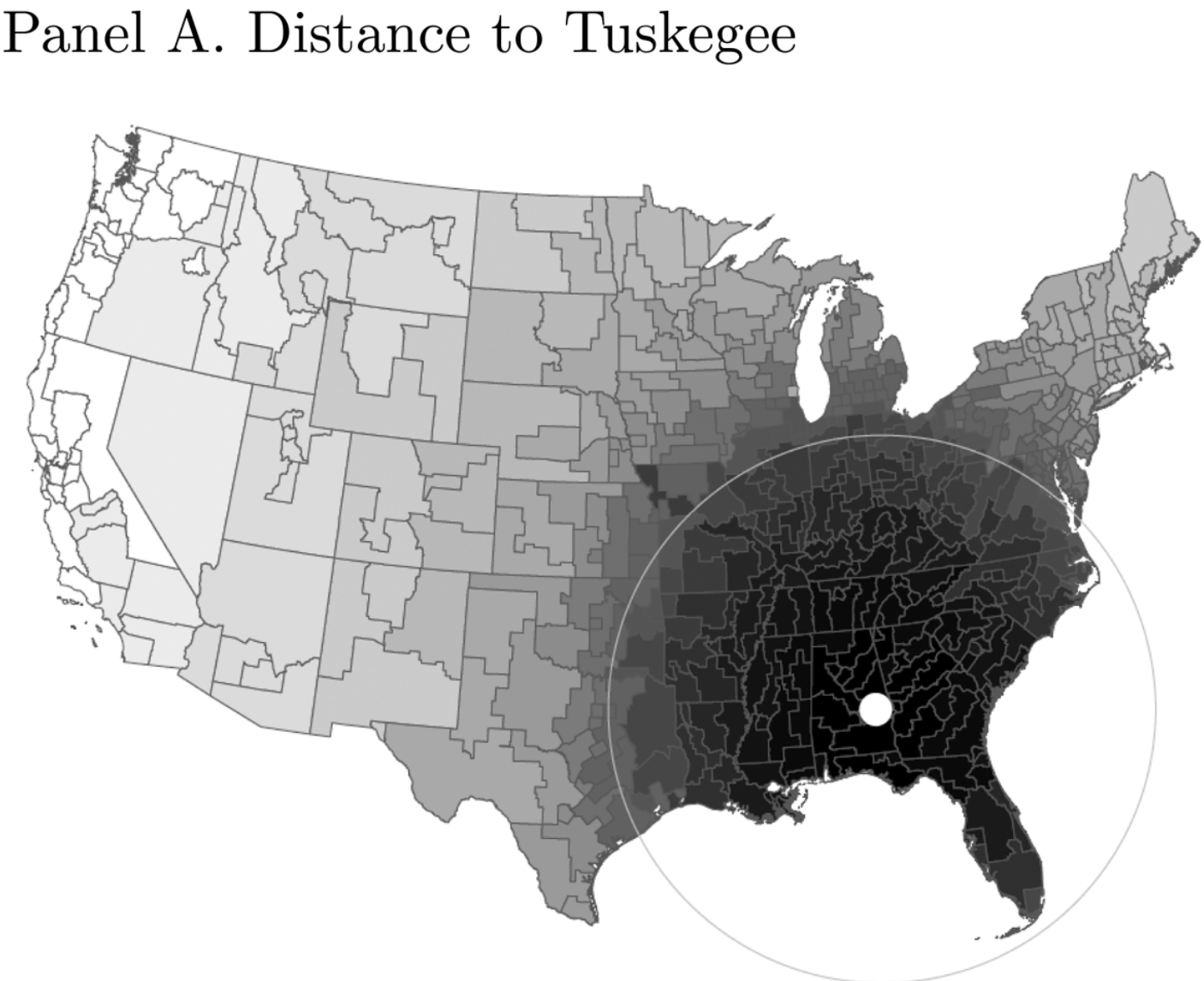
The Black-minus-white difference in mortality rates at ages 55–64, deaths per 1,000, for men (solid) and women (dashed), 1968–1988. The vertical line is the disclosure. Panel D, ages 65–74, shows the same break for men. Infant and child mortality, panels A and B, converge straight through 1972. Alsan and Wanamaker (2018), Figure I, panel C.

The data

- **Utilization.** National Health Interview Survey, 1969–1977, in the restricted version with state identifiers, used inside Census Research Data Centers. Nonveteran men and women aged 45–74.
- **Mortality.** CDC Compressed Mortality Files, 1968–1987, aggregated to state economic areas and two-year periods, ages 45–74, all causes and chronic causes. Rates are age-adjusted to the 1940 population so an aging population does not read as rising mortality.
- **Mistrust.** General Social Survey, 1998, the first year it asked whether doctors can be trusted. The sample is 801 respondents who were at least ten in 1972.
- **Supply controls.** Physicians, hospitals, beds, health centers, and public-health spending.

Alsan and Wanamaker (2018), pp. 420–423 and 446. The Public Health Service sponsored the survey. If mistrustful men refused it the utilization effects may be understated, though survey weights for Black men rose an insignificant 0.3 percent after 1972 (p. 421).

The treatment is proximity to Macon County



Distance from each state economic area to Macon County, Alabama. Darker is closer and the circle has a radius of about 1,000 kilometers. Proximity is the sample's maximum distance minus own distance, in thousands of kilometers. Alsan and Wanamaker (2018), Figure II, panel A.

Alsan and Wanamaker's specification

For person i of race r and sex g in state s and year t, where P_s is proximity to Macon County (the sample's maximum distance minus the state's distance, in thousands of kilometers), with individual controls X_{it} and fixed effects that absorb every lower-order interaction,

$$\begin{aligned} Y_{igrst} &= \alpha + \beta_1 \left(P_s \cdot \text{Post}_t \cdot \text{Black}_r \cdot \text{Male}_g\right) + \beta_2 \left(P_s \cdot \text{Post}_t \cdot \text{Male}_g\right) \\ &\quad + \beta_3 \left(P_s \cdot \text{Post}_t \cdot \text{Black}_r\right) + X_{it}\Gamma + \theta_{rgt} + \phi_{rgs} + \tau_{st} + \varepsilon_{igrst}. \end{aligned}$$

- θ_{rgt}, race by sex by year, absorbs anything national that hit Black men as a group, such as incarceration or unemployment trends.
- ϕ_{rgs}, race by sex by state, absorbs anything permanent about how Black men in Alabama differ from Black men in Ohio.
- τ_{st}, state by year, absorbs anything local that hit everyone, such as the Medicare and Medicaid rollouts.
- The remaining lower-order terms are absorbed. $P_s \cdot \text{Post}_t$ is in τ_{st}, $\text{Post}_t \cdot \text{Black}_r \cdot \text{Male}_g$ in θ_{rgt}, and $P_s \cdot \text{Black}_r \cdot \text{Male}_g$ in ϕ_{rgs}.

The identifying assumption, in their words. β_1 is the effect of the disclosure "provided there are no other systematic shocks to black men that affected healthcare utilization and mortality correlated with proximity to Macon County, but not due to the timing of the study's disclosure."

Alsan and Wanamaker (2018), equation (2), pp. 430–431.

The design in cells

Change after 1972, as a slope in proximity P, for each group.

	Men	**Women**
Black	$(\beta_1 + \beta_2 + \beta_3)P$	$\beta_3 P$
White	$\beta_2 P$	0

- Men minus women among Black people gives $(\beta_1 + \beta_2)P$. Men minus women among white people gives $\beta_2 P$. The difference is $\beta_1 P$.
- β_1 is the extra slope for Black men. β_2 is a slope shared by all men near Alabama and β_3 one shared by all Black people near Alabama. White men and Black women are the two control groups that remove them. White women are the reference.

With Table I's visit numbers, per thousand kilometers,

$$\underbrace{(-1.328 - 0.022 - 0.052)}_{\text{Black men}} - \underbrace{(-0.052)}_{\text{Black women}} - \underbrace{(-0.022)}_{\text{white men}} + \underbrace{0}_{\text{white women}} = -1.328.$$

The worked line reproduces β_1 by construction. Week 2 taught the difference in differences with Ravallion's example.

The baseline estimates

	Outpatient visits	Any visit	Log all-cause mortality	Log chronic mortality
Black-male mean before 1973	4.10	0.62	3.42	3.14
P· post · black · male	−1.328	−0.037	0.071	0.087
	(0.354)	(0.007)	(0.023)	(0.022)
P· post · male	−0.022	0.004	0.005	0.008
	(0.080)	(0.005)	(0.003)	(0.003)
P· post · black	−0.052	0.011	−0.017	−0.018
	(0.110)	(0.006)	(0.019)	(0.020)
Observations	220,954	220,954	17,737	17,611
Clusters	49	49	465	465

β_1 is large and precise in all four columns. β_2 and β_3 are far smaller.

Alsan and Wanamaker (2018), Table I, columns 1, 2, 5, and 7. Standard errors are clustered by state for utilization and by state economic area for mortality. P is in thousands of kilometers.

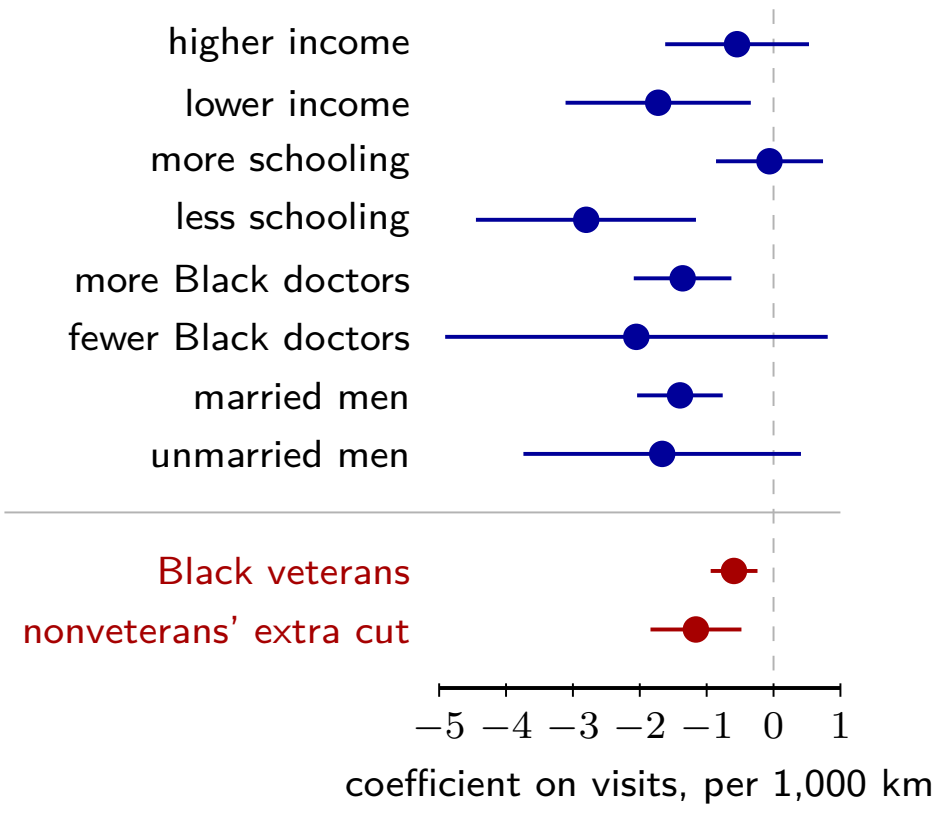
Magnitudes

One standard deviation of proximity to Macon County, after 1972, for Black men relative to the two control groups.

- **Visits.** 0.9 fewer outpatient contacts a year, 22 percent of the pre-disclosure mean of 4.1, and a 2.5-percentage-point lower chance of any visit.
- **Hospitals.** Admission down 1 percentage point, imprecise. Stays half a night longer, which the authors read as men arriving sicker.
- **Mortality.** Up 3.7 log points from all causes and 4.5 from chronic causes, or 1.1 and 0.8 deaths per 1,000. Deaths from accidents and violence do not move.
- **Life expectancy.** At age 45 Black men in the sample could expect 23.7 more years. Rebuilding the life table without the β_1 effect gives 25.2. The difference of up to 1.5 years is about 35 percent of the 1980 gap between Black and white men and 25 percent of the gap between Black men and Black women.

Alsan and Wanamaker (2018), pp. 435–438 and 449–450. The life table sets the effect to zero at the place farthest from Macon County, which is why the abstract says "up to" 1.5 years.

Who responded



- The poor and less educated responded most. The study's victims shared that profile.
- Effects were weaker, though imprecisely estimated, where Black doctors were more prevalent in 1970.
- Married Black women, the paper's proxy for experience of childbirth, kept converging. Unmarried Black women moved like men at a third the size.
- Black veterans cut 0.59 visits a year per thousand kilometers. Nonveterans cut 1.16 visits more, about 1.75 in all.

The pattern fits the belief frame, though the physician split and the estimate for unmarried men are imprecise.

Alsan and Wanamaker (2018), Tables II and V, pp. 438–443 and 448. Intervals are ± 1.96 standard errors. Blue rows are β_1 by subsample (Table II). Red rows are from the all-male sample (Table V), Black veterans against white veterans and the extra response of Black nonveterans.

Three answers at three levels

Life expectancy at birth rose by about 30 years between 1850 and 1950. What produced the gain and what happens when one input to it is destroyed?

	Input	**Mechanism**
Troesken	coercion, capital	constitutional constraints
Ferrie and Troesken	capital	typhoid and its sequelae
Alsan and Wanamaker	trust	a belief shock

Troesken, pp. 12–13. Epidemiology "is as much an exercise in political economy as it is in estimating statistical correlations," and diseases "are as much the result of choices, both public and private, as they are of geography and climate."

- **Democracies and disease.** Troesken's answer is about the American constitutional order more than democracy as such. It was better at financing waterworks than at compelling vaccination.
- **Trust and uptake.** Trust mattered enough to lower life expectancy at 45 for Black men in the sample by up to 1.5 years, net of supply-side controls. Week 1 previewed this paper for that reason.

Policy implications of the three papers

- **Water or cheaper programs?** Ferrie and Troesken compared Chicago in 1880 with the developing world of the 2000s. Their return on waterworks says the expensive investment paid. Why Chicago's works kept lagging its growth is an open question.

- **Why demand for proven health products is low.** Development economists call it the uptake paradox. A reading consistent with Alsan and Wanamaker is that a history of exploitation lowers demand for the service the exploiting institution supplies and that the response is rational while the exploitation continues.

- **Coercion has a cost.** Troesken's third reason. The capacity that vaccinates a population can be turned on a minority. In *Buck v. Bell* (1927) the Supreme Court upheld compulsory sterilization, citing the compulsory-vaccination case *Jacobson v. Massachusetts* (1905).

Troesken (2015), pp. 5–6. *Buck v. Bell*, 274 U.S. 200 (1927). These are open questions for class.

Next week

- **Week 9, 2 November. Infrastructure.**

- Hornbeck and Keniston on the Great Boston Fire of November 1872, which destroyed 776 buildings across 65 acres and raised land values on burned plots and nearby unburned ones.

- Hornbeck and Rotemberg on how much the railroads raised aggregate productivity in an economy with distorted input markets.

Troesken counts waterworks among the largest public investments in American history.